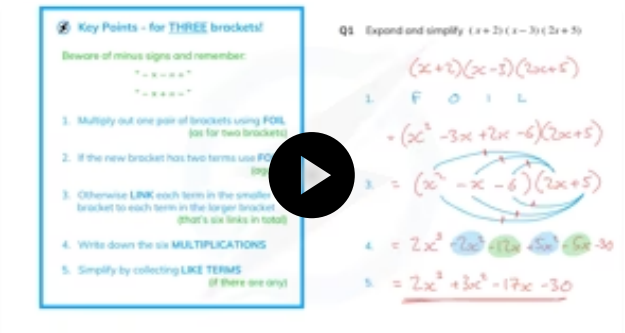
Expanding Triple Brackets

Exam code: 0607

Download PDF

Written by: Mark Curtis
Reviewed by: Dan Finlay
Updated on 20 January 2026



Did this video help you? Yes No

Expanding three brackets

How do I expand three brackets?

- Multiply out any **two of the brackets** using a standard method and **simplify**
- Then multiply the resulting expression by the third (unused) bracket
- This step often looks like $(x + a)(x^2 + bx + c)$
- Every term in the first bracket must be multiplied with every term in the second bracket
- A **grid** can help to keep track of all the terms
  - E.g. $(x + 2)(x^2 + 3x + 1)$

	x^2	$+3x$	$+1$
x	x^3	$3x^2$	x
$+2$	$2x^2$	$6x$	2

- Add all the terms inside the grid together
  - $x^3 + 2x^2 + 3x^2 + 6x + x + 2$
- **Simplify** by collecting any like terms
  - $x^3 + 5x^2 + 7x + 2$

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Worked Example

Expand $(2x-3)(x+4)(3x-1)$.

Answer:

Expand and simplify the first two brackets, for example using the FOIL method

$$\begin{aligned}&(2x-3)(x+4)\\&=2x\times x+2x\times 4+(-3)\times x+(-3)\times 4\\&=2x^2+8x-3x-12\\&=2x^2+5x-12\end{aligned}$$

Rewrite the original expression with the first two brackets expanded

$$(2x^2+5x-12)(3x-1)$$

Multiply all of the terms in the first set of brackets by all of the terms in the second set of brackets

A grid can help when there are many terms to multiply together (e.g. write $2x^2+5x-12$ in the vertical column and $3x-1$ in the horizontal column, then multiply corresponding terms)

	$3x$	-1
$2x^2$	$6x^3$	$-2x^2$
$5x$	$15x^2$	$-5x$
-12	$-36x$	12

Write out the multiplied terms

$$6x^3-2x^2+15x^2-5x-36x+12$$

Collect the like terms to simplify

$$\mathbf{6x^3+13x^2-41x+12}$$

Teacher only

Test your students on this in just a few clicks

- Create exam-aligned tests
- Tailor format & difficulty
- Quizzes or homework in minutes

Try Test Builder →

Test yourself

Was this revision note helpful? Yes No

Resources

Learning Hub
Scholarship
Join
Past Papers
Solution Banks
Sitemap

Dashboard

Overview
Brand ambassador
Profile
Subscription
Settings
Log out

Company

About us
Exam Specificity
Content Quality
Promotions
Jobs
Terms
Privacy
Cookie Policy
Help and Support

Subjects

Biology
Chemistry
Physics
Maths
Geography
English Literature
Psychology
All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.
Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.